SOLVED PRACTICE WORKBOOK

Semiconductor Mission and Electronics Manufacturing - Solved Practice Workbook

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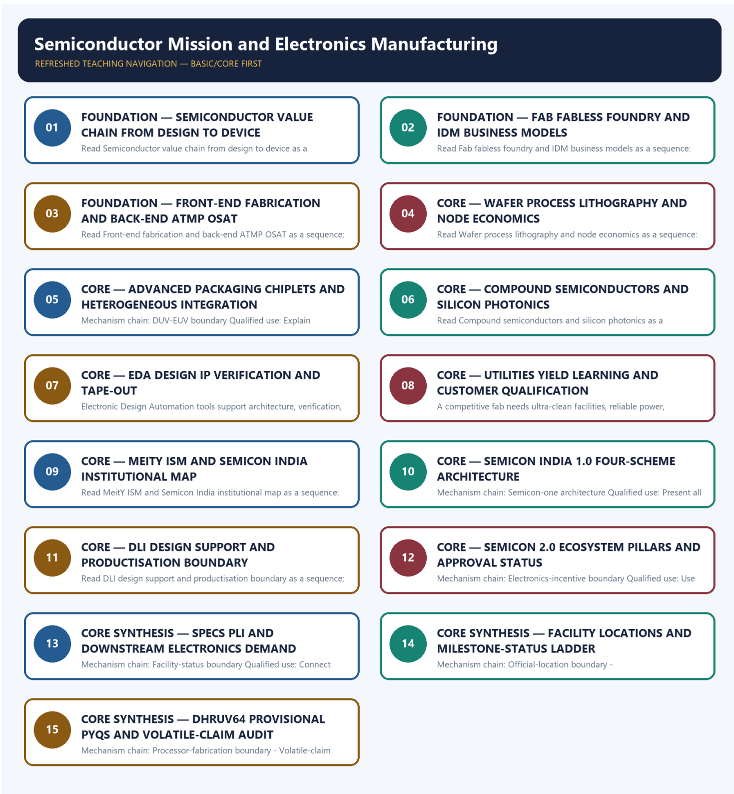

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CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Semiconductor Mission and Electronics Manufacturing - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Value-chain boundary?

A. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. B. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. C. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. D. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Answer: A. Explanation: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Value-chain boundary?

A. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. B. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. C. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. D. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical.

Answer: B. Explanation: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Value-chain boundary without changing its institution, unit or status?

A. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. B. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. C. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. D. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications.

Answer: C. Explanation: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Value-chain boundary?

A. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. B. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. C. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. D. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.

Answer: D. Explanation: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Fabless-foundry-IDM boundary?

A. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. B. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. C. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. D. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications.

Answer: A. Explanation: A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Fabless-foundry-IDM boundary?

A. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. B. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. C. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. D. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications.

Answer: B. Explanation: A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Fabless-foundry-IDM boundary without changing its institution, unit or status?

A. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. B. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. C. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. D. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.

Answer: C. Explanation: A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Fabless-foundry-IDM boundary?

A. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. B. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym.

Answer: D. Explanation: A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Front-end-back-end boundary?

A. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. B. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. C. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. D. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical.

Answer: A. Explanation: Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Front-end-back-end boundary?

A. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. B. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications.

Answer: B. Explanation: Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Front-end-back-end boundary without changing its institution, unit or status?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. C. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. D. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source.

Answer: C. Explanation: Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Front-end-back-end boundary?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Answer: D. Explanation: Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies ATMP-OSAT boundary?

A. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. B. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source.

Answer: A. Explanation: ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of ATMP-OSAT boundary?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source.

Answer: B. Explanation: ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses ATMP-OSAT boundary without changing its institution, unit or status?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. C. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. D. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.

Answer: C. Explanation: ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about ATMP-OSAT boundary?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. C. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. D. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical.

Answer: D. Explanation: ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Process-node boundary?

A. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. B. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. C. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. D. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.

Answer: A. Explanation: A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Process-node boundary?

A. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. B. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor.

Answer: B. Explanation: A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Process-node boundary without changing its institution, unit or status?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. C. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. D. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.

Answer: C. Explanation: A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Process-node boundary?

A. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. B. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. C. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. D. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications.

Answer: D. Explanation: A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies DUV-EUV boundary?

A. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. B. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. C. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. D. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.

Answer: A. Explanation: Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of DUV-EUV boundary?

A. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. B. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. C. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. D. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.

Answer: B. Explanation: Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses DUV-EUV boundary without changing its institution, unit or status?

A. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. B. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. C. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. D. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.

Answer: C. Explanation: Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about DUV-EUV boundary?

A. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. B. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. C. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. D. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source.

Answer: D. Explanation: Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Advanced-packaging boundary?

A. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. B. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. C. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. D. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.

Answer: A. Explanation: Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Advanced-packaging boundary?

A. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. B. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. C. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. D. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives.

Answer: B. Explanation: Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Advanced-packaging boundary without changing its institution, unit or status?

A. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. B. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. C. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. D. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives.

Answer: C. Explanation: Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Advanced-packaging boundary?

A. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. B. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. C. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. D. Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.

Answer: D. Explanation: Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Compound-photonics boundary?

A. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. B. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. C. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. D. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.

Answer: A. Explanation: Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Compound-photonics boundary?

A. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. B. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. C. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. D. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.

Answer: B. Explanation: Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Compound-photonics boundary without changing its institution, unit or status?

A. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. B. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. C. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. D. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support.

Answer: C. Explanation: Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Compound-photonics boundary?

A. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. B. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. C. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. D. Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor.

Answer: D. Explanation: Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies EDA-design-IP boundary?

A. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. B. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. C. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. D. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.

Answer: A. Explanation: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of EDA-design-IP boundary?

A. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. B. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. C. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. D. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support.

Answer: B. Explanation: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses EDA-design-IP boundary without changing its institution, unit or status?

A. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. D. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance.

Answer: C. Explanation: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about EDA-design-IP boundary?

A. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. D. Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.

Answer: D. Explanation: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Utility-yield boundary?

A. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. B. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. C. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. D. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives.

Answer: A. Explanation: A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Utility-yield boundary?

A. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. B. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. C. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. D. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support.

Answer: B. Explanation: A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Utility-yield boundary without changing its institution, unit or status?

A. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. D. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Answer: C. Explanation: A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Utility-yield boundary?

A. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. B. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. C. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. D. A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.

Answer: D. Explanation: A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies ISM-institution boundary?

A. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. D. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance.

Answer: A. Explanation: India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of ISM-institution boundary?

A. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. B. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. C. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. D. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Answer: B. Explanation: India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses ISM-institution boundary without changing its institution, unit or status?

A. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. B. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. C. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. D. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Answer: C. Explanation: India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about ISM-institution boundary?

A. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. B. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. C. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. D. India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives.

Answer: D. Explanation: India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Semicon-one architecture?

A. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. D. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance.

Answer: A. Explanation: Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Semicon-one architecture?

A. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. B. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. C. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. D. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent.

Answer: B. Explanation: Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Semicon-one architecture without changing its institution, unit or status?

A. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. B. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. C. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. D. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.

Answer: C. Explanation: Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Semicon-one architecture?

A. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. B. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. C. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. D. Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support.

Answer: D. Explanation: Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies DLI-boundary?

A. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. D. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.

Answer: A. Explanation: Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of DLI-boundary?

A. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. B. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. C. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. D. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Answer: B. Explanation: Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses DLI-boundary without changing its institution, unit or status?

A. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. B. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. C. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. D. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.

Answer: C. Explanation: Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about DLI-boundary?

A. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. B. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. C. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. D. Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance.

Answer: D. Explanation: Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Semicon-two boundary?

A. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. B. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. C. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. D. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Answer: A. Explanation: Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Semicon-two boundary?

A. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. B. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. C. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. D. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat.

Answer: B. Explanation: Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Semicon-two boundary without changing its institution, unit or status?

A. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. B. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. C. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. D. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat.

Answer: C. Explanation: Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Semicon-two boundary?

A. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. B. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. C. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. D. Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent.

Answer: D. Explanation: Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Electronics-incentive boundary?

A. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. B. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. C. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. D. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.

Answer: A. Explanation: SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Electronics-incentive boundary?

A. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. B. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. C. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. D. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale.

Answer: B. Explanation: SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Electronics-incentive boundary without changing its institution, unit or status?

A. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. B. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. C. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. D. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions.

Answer: C. Explanation: SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Electronics-incentive boundary?

A. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. B. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. C. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. D. SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Answer: D. Explanation: SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Facility-status boundary?

A. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. B. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. C. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. D. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability.

Answer: A. Explanation: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Facility-status boundary?

A. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. B. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. C. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. D. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability.

Answer: B. Explanation: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Facility-status boundary without changing its institution, unit or status?

A. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. B. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. C. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. D. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.

Answer: C. Explanation: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Facility-status boundary?

A. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. B. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. C. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. D. Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.

Answer: D. Explanation: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Official-location boundary?

A. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. B. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. C. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. D. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions.

Answer: A. Explanation: Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Official-location boundary?

A. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. B. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. C. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. D. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability.

Answer: B. Explanation: Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Official-location boundary without changing its institution, unit or status?

A. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. B. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. C. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. D. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.

Answer: C. Explanation: Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Official-location boundary?

A. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. B. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. C. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. D. Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat.

Answer: D. Explanation: Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Commissioning-evidence boundary?

A. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. B. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. C. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. D. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability.

Answer: A. Explanation: Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Commissioning-evidence boundary?

A. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. B. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. C. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. D. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.

Answer: B. Explanation: Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Commissioning-evidence boundary without changing its institution, unit or status?

A. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. B. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. C. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. D. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Answer: C. Explanation: Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Commissioning-evidence boundary?

A. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. B. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. C. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. D. Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale.

Answer: D. Explanation: Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Processor-fabrication boundary?

A. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. B. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. C. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. D. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym.

Answer: A. Explanation: The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Processor-fabrication boundary?

A. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. B. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. C. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. D. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym.

Answer: B. Explanation: The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Processor-fabrication boundary without changing its institution, unit or status?

A. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. B. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. C. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. D. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Answer: C. Explanation: The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Processor-fabrication boundary?

A. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. B. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. C. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. D. The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability.

Answer: D. Explanation: The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile-claim boundary?

A. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. B. The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. C. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. D. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Answer: A. Explanation: Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile-claim boundary?

A. A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. B. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. C. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. D. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical.

Answer: B. Explanation: Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile-claim boundary without changing its institution, unit or status?

A. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. B. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. C. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. D. Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Answer: C. Explanation: Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile-claim boundary?

A. A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. B. Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. C. ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. D. Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions.

Answer: D. Explanation: Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2025 GS-III semiconductor-challenges and India Semiconductor Mission demand plus provisional 2026 processor and plant-location concepts here. The cards preserve official verbs and do not invent an answer letter, project total, node size or commercial-production claim.

PYQ DEMAND CARD 1 - 2025 GS-III

Demand: Analyse challenges faced by India's semiconductor industry and mention the salient features of the India Semiconductor Mission.

Status: Audited routed Mains demand; the card covers the complete value chain, ISM architecture and implementation constraints without treating later developments as part of the original official model answer.

Model solution: Value-chain boundary: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **DUV-EUV boundary:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Utility-yield boundary:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **ISM-institution boundary:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Semicon-one architecture:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **DLI-boundary:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Semicon-two boundary:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Facility-status boundary:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2025 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Value-chain boundary: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **DUV-EUV boundary:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Utility-yield boundary:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **ISM-institution boundary:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Semicon-one architecture:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **DLI-boundary:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Semicon-two boundary:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Facility-status boundary:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Analyse challenges faced by India's semiconductor industry and mention the salient features of the India Semiconductor Mission. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Audited routed Mains demand; the card covers the complete value chain, ISM architecture and implementation constraints without treating later developments as part of the original official model answer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Value-chain boundary: The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **DUV-EUV boundary:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Utility-yield boundary:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **ISM-institution boundary:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Semicon-one architecture:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **DLI-boundary:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Semicon-two boundary:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Facility-status**

boundary: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2025 GS-III”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 2 - 2026 Prelims GS-I

Demand: Assess DHRUV64 processor, DIR-V programme and indigenous computing-capability statements.

Status: Provisional routed objective concept; DHRUV64's audited processor description is retained, but the unverified DIR-V ordinal proposition and answer letter are not asserted.

Model solution: EDA-design-IP boundary: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Processor-fabrication boundary:** The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 2 - 2026 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: EDA-design-IP boundary: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Processor-fabrication boundary:** The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess DHRUV64 processor, DIR-V programme and indigenous computing-capability statements. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Provisional routed objective concept; DHRUV64's audited processor description is retained, but the unverified DIR-V ordinal proposition and answer letter are not asserted. **Analysis:**

Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: EDA-design-IP boundary: Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Processor-fabrication boundary:** The audited owner identifies DHRUV64 as a homegrown 1.0-GHz, 64-bit dual-core microprocessor linked to India's RISC-V and processor-IP effort, but a processor design headline does not prove domestic fabrication, volume production or deployed capability. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2026 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2026 Prelims GS-I

Demand: Match Indian semiconductor projects or plants with their announced state-level locations.

Status: Provisional routed objective concept; official location and status distinctions are taught without inferring an answer letter, aggregate project count or commercial production.

Model solution: Facility-status boundary: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Official-location boundary:** Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. **Commissioning-evidence boundary:** Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 3 - 2026 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Facility-status boundary: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Official-location boundary:** Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. **Commissioning-evidence boundary:** Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Match Indian semiconductor projects or plants with their announced state-level locations. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Provisional routed objective concept; official location and status distinctions are taught without inferring an answer letter, aggregate project count or commercial production. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Facility-status boundary: Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Official-location boundary:** Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. **Commissioning-evidence boundary:** Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. **Volatile-claim boundary:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 3 - 2026 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish semiconductor design, wafer fabrication and ATMP or OSAT. Answer in about 150 words.

Model thesis: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Front-end-back-end boundary. **Named evidence/example:** Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATMP-OSAT boundary. **Named evidence/example:** ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.
- Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.
- ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical.
- Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.

Qualified conclusion: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Front-end-back-end boundary. **Named evidence/example:** Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATMP-OSAT boundary. **Named evidence/example:** ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on “Distinguish semiconductor design, wafer fabrication and ATMP or OSAT. Answer in about 150…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Front-end-back-end boundary. **Named evidence/example:** Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATMP-OSAT boundary. **Named evidence/example:** ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Front-end-back-end boundary. **Named evidence/example:** Wafer fabrication is the front-end creation of transistor and interconnect layers, while ATMP or OSAT is the back-end assembly, testing, marking and packaging of diced dies; packaging capacity is not wafer-fabrication capacity.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ATMP-OSAT boundary. **Named evidence/example:** ATMP names assembly, testing, marking and packaging process steps, whereas OSAT names the outsourced service model that performs such back-end work; the terms overlap operationally but are not identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Distinguish semiconductor design, wafer fabrication and ATMP or OSAT. Answer in about 150…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate a fab, foundry, fabless company and integrated device manufacturer. Answer in about 150 words.

Model thesis: Claim: Fabless-foundry-IDM boundary. **Named evidence/example:** A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym.

Qualified conclusion: Claim: Fabless-foundry-IDM boundary. **Named evidence/example:** A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on “Differentiate a fab, foundry, fabless company and integrated device manufacturer. Answer in…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Fabless-foundry-IDM boundary. **Named evidence/example:** A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Fabless-foundry-IDM boundary. **Named evidence/example:** A fabless firm designs chips and outsources wafer manufacture, a foundry manufactures to customers' designs, and an integrated device manufacturer designs and fabricates its own products; a fab is the plant and is not itself a business-model synonym. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Differentiate a fab, foundry, fabless company and integrated device manufacturer. Answer in…", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain why advanced packaging and compound semiconductors can form a strategic capability route for India. Answer in about 250 words.

Model thesis: Claim: Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.
- Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor.
- A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.

Qualified conclusion: Claim: Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain why advanced packaging and compound semiconductors can form a strategic capability…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** Connect mechanism -> named

system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain why advanced packaging and compound semiconductors can form a strategic capability…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Describe the institutional and scheme architecture of India's semiconductor mission. Answer in about 250 words.

Model thesis: Claim: ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-one architecture. **Named evidence/example:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DLI-boundary. **Named evidence/example:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-two boundary. **Named evidence/example:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Electronics-incentive boundary. **Named evidence/example:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives.
- Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support.
- Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance.
- Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent.
- SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.

Qualified conclusion: Claim: ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-one architecture. **Named evidence/example:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DLI-boundary. **Named evidence/example:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-two boundary. **Named evidence/example:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Electronics-incentive boundary. **Named evidence/example:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **describe** requires a direct position on “Describe the institutional and scheme architecture of India's semiconductor mission. Answer…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-one architecture. **Named evidence/example:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DLI-boundary. **Named evidence/example:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-two boundary. **Named evidence/example:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Electronics-incentive boundary. **Named evidence/example:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-one architecture. **Named evidence/example:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DLI-boundary. **Named evidence/example:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-two boundary. **Named evidence/example:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Electronics-incentive boundary. **Named evidence/example:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Describe the institutional and scheme architecture of India's semiconductor mission. Answer…", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Analyse the challenges faced by India's semiconductor industry and the response of the India Semiconductor Mission. Answer in about 300 words.

Model thesis: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DUV-EUV boundary. **Named evidence/example:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-one architecture. **Named evidence/example:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DLI-boundary. **Named evidence/example:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-two boundary. **Named evidence/example:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Facility-status boundary. **Named evidence/example:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.
- Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source.
- Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies.
- A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.
- India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives.
- Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support.
- Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance.
- Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent.
- Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.
- Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions.

Qualified conclusion: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DUV-EUV boundary. **Named evidence/example:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-one architecture. **Named evidence/example:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DLI-boundary. **Named evidence/example:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Semicon-two boundary. **Named evidence/example:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Facility-status boundary. **Named evidence/example:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **analyse** requires a direct position on “Analyse the challenges faced by India's semiconductor industry and the response of the India…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DUV-EUV boundary. **Named evidence/example:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** ISM-institution boundary. **Named evidence/example:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; 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Analytical body:

1. **Claim and named evidence:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; design capability can exist without domestic wafer fabrication and still retain external tool-chain dependencies. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** India Semiconductor Mission is the MeitY-linked implementing agency for semiconductor and display ecosystem schemes; it is distinct from the Semicon India scheme package and from separate electronics-manufacturing incentives. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Semicon India 1.0 has a Rs 76,000 crore umbrella covering semiconductor fabs, display fabs, compound-semiconductor or silicon-photonics or sensor fabs plus ATMP-OSAT, and Design Linked Incentive support. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Design Linked Incentive supports domestic chip design, design infrastructure, startups and productisation; it is not a direct wafer-fab subsidy and does not establish fabrication self-reliance. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Semicon 2.0 was approved by the Union Cabinet on 15 July 2026 with a Rs 1,27,500 crore outlay and an official ecosystem emphasis spanning design, machines and materials, more fabs, ATMP or OSAT, research and development, and talent. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
10. **Claim and named evidence:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. **Analysis:**

Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DUV-EUV boundary. **Named evidence/example:** Deep-ultraviolet lithography supports mature and mid-range manufacturing, while extreme-ultraviolet lithography is a leading-edge chokepoint concentrated in one global supplier and exposed to export controls; a smaller-node claim requires an explicit official source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** EDA-design-IP boundary. **Named evidence/example:** Electronic Design Automation tools support architecture, verification, layout and tape-out, while reusable design intellectual property supplies circuit blocks; 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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Analyse the challenges faced by India's semiconductor industry and the response of the India…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Evaluate India's semiconductor strategy through value-chain depth, facility status, electronics demand and selective resilience. Answer in about 300 words.

Model thesis: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Process-node boundary. **Named evidence/example:** A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Electronics-incentive boundary. **Named evidence/example:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Facility-status boundary. **Named evidence/example:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; 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these verbs do not establish node, yield or commercial-scale sale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage.
- A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications.
- Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step.
- Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor.
- A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes.
- SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme.
- Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one.
- Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat.
- Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale.
- Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions.

Qualified conclusion: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Process-node boundary. **Named evidence/example:** A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; 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Demand decoding: The directive **evaluate** requires a direct position on “Evaluate India's semiconductor strategy through value-chain depth, facility status,…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Process-node boundary. **Named evidence/example:** A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Electronics-incentive boundary. **Named evidence/example:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Facility-status boundary. **Named evidence/example:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Official-location boundary. **Named evidence/example:** Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Commissioning-evidence boundary. **Named evidence/example:** Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs do not establish node, yield or commercial-scale sale. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; inauguration alone proves none of these operating outcomes. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** SPECS and electronics or IT-hardware PLI instruments complement semiconductor policy through component and downstream manufacturing incentives, but they are separate from ISM and cannot be merged into one scheme. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Official semiconductor evidence must preserve the ladder of approval, fiscal-support agreement, groundbreaking, inauguration, demonstrated output, customer qualification and commercial supply; no earlier rung proves a later one. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Official 2024 material identifies a Dholera fab and OSAT or ATMP units at Morigaon and Sanand, while the 5 May 2026 official update identifies Crystal Matrix's integrated compound-semiconductor fabrication plus ATMP project in Dholera and Suchi Semicon's OSAT unit in Surat. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** Official records identify Micron ATMP at Sanand as inaugurated on 28 February 2026, Kaynes Semicon at Sanand on 31 March 2026 and CG Semi OSAT at Sanand on 4 July 2026; these verbs

do not establish node, yield or commercial-scale sale. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

10. **Claim and named evidence:** Project counts, node sizes, capacities, yields, investment realised, commissioning, commercial output and event outcomes require a dated official source; the provisional 2026 questions and keys supply demands, not answer letters or independent proof of their propositions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Value-chain boundary. **Named evidence/example:** The semiconductor chain runs from design and intellectual property through wafer fabrication, assembly, testing and packaging to boards, modules and finished electronics; strength at one stage does not establish control of every stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Process-node boundary. **Named evidence/example:** A process node is now largely a generation label associated with density and performance rather than a literal transistor dimension; mature nodes remain strategically useful for automotive, industrial, power and defence applications. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Advanced-packaging boundary. **Named evidence/example:** Chiplets, 2.5D and 3D integration, through-silicon vias and heterogeneous integration can improve system performance without leading-edge lithography, making advanced packaging a capability route rather than a trivial final assembly step. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Compound-photonics boundary. **Named evidence/example:** Compound semiconductors such as GaN, GaAs, InP and SiC combine multiple elements for power, radio-frequency or optoelectronic uses, while silicon photonics is a silicon platform for optical interconnects and is not a compound semiconductor. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Utility-yield boundary. **Named evidence/example:** A competitive fab needs ultra-clean facilities, reliable power, ultra-pure water, specialty gases and chemicals, process control, sustained yield learning and customer qualification; 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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Evaluate India's semiconductor strategy through value-chain depth, facility status,…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.